

Civic Hygiene and Waste Management: A Community Awareness Initiative.

Submitted for Bachelor of Science in Information Technology by,

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[student's declaration]

Acknowledgement

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My heartiest thanks go to the local residents, shopkeepers, and municipal workers of the survey area. Without their transparency and willingness to participate in interviews and surveys, this report would lack the necessary ground-level data. Their cooperation has provided me with a realistic perspective on the hygiene hurdles faced by our society.

Lastly, I would like to thank my parents and friends for their continuous moral support and motivation during the field visits and the documentation process of this project.

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SYIT

Abstract

"The rapid urbanization of cities like Mumbai has led to a massive increase in waste generation, making **Civic Hygiene and Waste Management** one of the most pressing challenges of the 21st century. This Community Engagement Project (CEP) is an initiative to analyze the existing sanitation infrastructure and public behavioral patterns within a specific urban locality. The primary objective of this study is to bridge the gap between theoretical environmental policies and their practical implementation at the grassroots level.

The research was conducted using a qualitative and quantitative approach, including field audits of 'black spots,' drainage inspections, and community surveys. The study reveals that while the municipal collection system is functional, factors such as lack of source segregation, improper disposal of single-use plastics, and irregular cleaning of public bins contribute significantly to local hygiene degradation. The report further investigates the impact of these practices on the local ecosystem, specifically focusing on the potential for drainage clogging and the spread of vector-borne diseases.

Through this project, an attempt has been made to promote the '3R' (Reduce, Reuse, Recycle) framework and the 'Swachh Bharat' mission. The findings suggest that long-term sustainability can only be achieved through active community participation and a shift in civic consciousness. This abstract serves as a comprehensive summary of the project's journey—from identifying hygiene bottlenecks to proposing scalable solutions for a cleaner and healthier living environment. The conclusion of this report emphasizes that technology and policy must be supported by individual responsibility to ensure a sustainable future for all urban dwellers."

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1.INTRODUCTION

In the modern era of rapid urbanization and global population expansion, the maintenance of civic hygiene and efficient waste management has emerged as one of the most critical challenges facing metropolitan governance. Civic hygiene is defined as the collective responsibility of a community to maintain sanitary conditions in public spaces such as streets, marketplaces, and residential corridors. In a massive urban landscape like Mumbai, the sheer volume of solid waste generated daily—ranging from organic household materials to complex non-biodegradable polymers—demands a highly organized and conscious disposal system. While national initiatives like the "Swachh Bharat Abhiyan" (Clean India Mission) have provided a strong foundation for a cleaner India, the ultimate success of such policies is determined by grassroots-level participation. Without active community engagement, even the most advanced waste management infrastructures fail to achieve their intended health and environmental outcomes.

This field project, titled "**Civic Hygiene and Waste Management: A Community Awareness Initiative,**" is designed to bridge the gap between academic learning and social responsibility. The primary objective is to conduct a localized hygiene audit to identify "black spots" where waste accumulation is high and to evaluate the community's current understanding of waste segregation. By interacting with local residents and shopkeepers, this initiative aims to promote the vital habit of separating "Dry Waste" from "Wet Waste" at the source. As students of Information Technology, we recognize that while technical solutions can improve waste tracking, the core of a sustainable and disease-free society lies in human awareness and behavioral change. This report documents our field observations and provides actionable recommendations to enhance the hygiene standards of our locality, ensuring a healthier and more sustainable environment for all. **Objectives of the Study**

The primary objective of this field project is to assess the current state of public sanitation and waste management within a specific urban locality. By engaging directly with the community, this study aims to identify the most common types of waste generated in residential and commercial zones and evaluate the effectiveness of existing waste collection mechanisms. Furthermore, it aims to spread awareness regarding the hazards of improper waste dumping and the benefits of the "3Rs" (Reduce, Reuse, Recycle). This ensures that citizens are encouraged to adopt responsible habits like using separate bins for dry and wet waste.

This study is focused specifically on a selected neighborhood to provide a detailed, grassroots-level analysis of civic hygiene. While the observations are localized, they reflect the broader challenges faced by metropolitan areas like Mumbai. The project relies on direct field observations, community interactions, and visual documentation to draw conclusions. Due to time constraints, the study focuses on residential awareness rather than industrial waste management, ensuring that the recommendations provided are practical and easy for everyday citizens to implement.

2.OBJECTIVES OF THE STUDY

Understanding Waste Categorization One of the most important objectives is to educate the community on how to distinguish between different types of waste. Most households mix everything in one bag, which makes recycling impossible. This project aims to teach residents the difference between "Wet Waste" (food scraps, vegetable peels) and "Dry Waste" (plastic, paper, glass). By doing this, we reduce the total volume of trash that goes to the dumping grounds.

Elimination of Public Health Risks Improper waste disposal is the main reason for the spread of diseases like Malaria, Dengue, and Typhoid. This study aims to identify "black spots" in the locality—areas where garbage is dumped illegally. By identifying these spots, we can work toward a permanent cleanup to prevent these areas from becoming breeding grounds for insects and bacteria, eventually improving the health of every resident.

Reduction of Single-Use Plastic Plastic pollution is a major crisis for Mumbai's environment. An objective of this study is to observe how much single-use plastic (like thin bags and straws) is still being used despite the government ban. We aim to encourage shopkeepers and customers to switch to cloth bags or paper alternatives, ensuring that plastic doesn't end up blocking the city's drainage systems.

Community Engagement and Behavioral Change Cleaning a city is not just the job of the government; it is a collective responsibility. This project seeks to change the mindset of the people. Instead of a "someone else will clean it" attitude, we aim to inspire a sense of ownership. If people feel responsible for the street in front of their house, the entire neighborhood stays clean naturally.

Evaluation of Local Sanitation Infrastructure We need to check if the local area has enough dustbins and if the garbage collection trucks are coming on time every day. This objective involves auditing the available resources to see if they are sufficient for the number of people living in the area. If the bins are always overflowing, it shows that the locality needs more infrastructure or more frequent collection cycles.

Promoting Sustainable Living for the Future The final objective is to promote the "3R" principle: Reduce, Reuse, and Recycle. We want to show people that a lot of what they throw away can actually be reused or recycled into something useful. This helps in conserving natural resources and ensures a cleaner, greener environment for the next generation.

1. Primary Objective The fundamental goal of this project is to foster a sense of social responsibility among the residents of the local community regarding civic hygiene. This involves conducting a comprehensive field audit to understand how household and commercial waste is currently managed and identifying the specific barriers that prevent people from maintaining a clean and sustainable environment.

2. Secondary Objectives

- **Identification of Waste Streams:** To categorize the types of waste being generated in the locality, such as biodegradable kitchen waste, recyclable plastics, hazardous electronic waste, and non-recyclable sanitary waste, to better understand the pressure on local disposal systems.
- **Assessment of Disposal Habits:** To observe and record the existing disposal patterns of the community, checking whether waste is being dumped in open spaces, overflowing bins, or being handed over directly to municipal collection vehicles in a segregated manner.
- **Evaluation of Infrastructure Efficiency:** To analyze the availability and placement of public dustbins and the regularity of the municipal waste collection cycle, ensuring that the infrastructure provided by the local authorities is sufficient for the population density of the area.
- **Raising Public Awareness:** To educate residents, especially shopkeepers and street vendors, about the long-term environmental and health risks associated with plastic pollution and improper waste accumulation.
- **Promoting Source Segregation:** To demonstrate and explain the "Wet Waste" vs. "Dry Waste" segregation process, emphasizing how this simple step at home can drastically improve the efficiency of large-scale waste processing plants.
- **Mitigation of Health Hazards:** To identify "black spots" or illegal dumping sites that act as breeding grounds for mosquitoes and disease-carrying vectors, and to propose solutions for their permanent cleanup.
- **Encouraging Community Participation:** To inspire local youth and residents to take ownership of their surroundings through small-scale initiatives, ensuring that the habit of cleanliness extends beyond the duration of this field project.
- **Technological Integration and Monitoring:** To explore how modern reporting tools or digital awareness can help in tracking hygiene issues, allowing citizens to communicate more effectively with municipal bodies for faster grievance redressal.

3.METHODOLOGY (ACTION PLAN)

Study Area Selection For this field project, a specific residential locality in Mumbai was selected. The area consists of a mix of high-rise buildings and small-scale commercial shops. This location was chosen because it represents a typical urban setting where waste management challenges are most visible, especially during peak morning hours when municipal collection takes place.

Data Collection Techniques To gather accurate information for this report, the following methods were used:

- **Direct Field Observation:** I visited the locality at different times of the day to observe the cleanliness of the streets and the condition of public dustbins.
- **Photography:** Visual evidence was captured to document the "black spots" where garbage accumulates and to show the contrast between clean and neglected areas.
- **Community Interaction:** Brief informal discussions were held with local residents, shopkeepers, and Safai Karamcharis (cleaning staff) to understand their daily struggles with waste disposal.

Proposed Action Plan The project was executed in a structured manner over several days:

- **Phase 1 (Planning):** Selecting the topic and identifying the target area for the field study.
- **Phase 2 (Observation):** Walking through the streets to record the current hygiene status and identifying the types of waste (Plastic, organic, etc.).
- **Phase 3 (Awareness):** Interacting with people to explain the importance of dry and wet waste segregation.
- **Phase 4 (Documentation):** Compiling all the photos, notes, and observations into this final report.

Detailed Field Observations During the field visits, I focused on identifying the specific types of garbage found on the streets. I noticed that a large percentage of the waste consisted of discarded food packaging, plastic bottles, and organic kitchen waste. By walking through the narrow lanes and main roads of the locality, I was able to map out the areas that require urgent attention from the municipal authorities.

Tools Used for Data Recording To ensure the accuracy of the report, I used digital tools like a smartphone camera for high-resolution documentation and a notebook to record live feedback from the residents. Each observation was timestamped to track how hygiene levels change during different hours of the day, specifically before and after the arrival of the garbage collection van.

Selection of the Study Area For this field project, a specific urban residential and commercial locality in Mumbai was selected as the primary site for investigation. This area was chosen because it represents a high-density population zone where waste management challenges are most prominent. The diversity of the locality—comprising residential apartments, street vendors, and small business establishments—provides a comprehensive view of how different sectors contribute to urban waste and how they manage hygiene on a daily basis.

Primary Data Collection Methods To ensure the report is based on factual evidence, a multi-dimensional approach to data collection was adopted:

- **Direct Field Observation:** Multiple visits were conducted at different intervals of the day (morning, afternoon, and evening) to monitor the consistency of waste collection and the accumulation patterns of garbage in public spaces.
- **Visual Documentation:** High-resolution digital photographs were captured to document "black spots" (illegal dumping sites) and to compare them with areas that are well-maintained by the municipal authorities.
- **Community Interaction:** Brief, informal interviews were conducted with local residents, shopkeepers, and the municipal cleaning staff (Safai Karamcharis). These interactions helped in understanding the practical difficulties faced by the community regarding waste segregation and disposal timings.

Tools and Materials Used The study was supported by simple yet effective tools to record and analyze data:

- **Observation Logbook:** A dedicated diary was maintained to record live notes, timestamps of garbage collection, and specific hygiene issues noticed during the walks.
- **Digital Imaging Device:** A smartphone camera was used for geotagged photography, ensuring that the evidence provided in the report is authentic and site-specific.
- **Survey Questionnaires:** A set of basic questions was prepared to gauge the level of awareness among residents regarding the "3R" principle (Reduce, Reuse, Recycle) and the plastic ban.

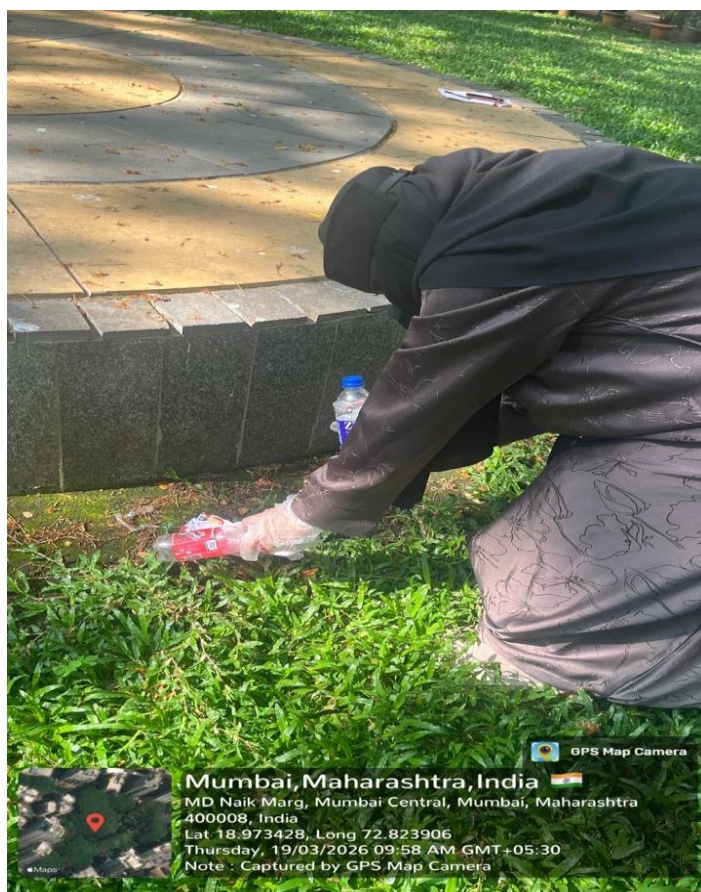
Step-by-Step Action Plan The project was executed following a structured timeline to ensure all objectives were met:

1. **Phase 1: Reconnaissance Survey:** An initial walkthrough was conducted to map out the placement of community dustbins and identify the main sources of waste.
2. **Phase 2: Detailed Audit:** A thorough inspection of the drainage systems and open plots was performed to check for clogging caused by non-biodegradable materials like plastic and thermocol.

3. **Phase 3: Awareness Drive:** During the field visits, residents were encouraged to practice source segregation (separating wet and dry waste) through verbal communication and demonstrations.
4. **Phase 4: Data Compilation:** All collected notes, photographs, and feedback were categorized and analyzed to form the basis of the "Observations and Analysis" chapter of this report.







4. Indicative List of Areas for CEP.

The scope of Community Engagement Projects (CEP) is vast and covers multiple dimensions of urban and rural life. The following areas represent the key sectors where students and volunteers can intervene to create a positive social impact:

4.1 Waste Management and Environmental Sanitation This is a primary area of focus for urban localities. It involves analyzing the lifecycle of waste from the point of generation to its final disposal.

- **Source Segregation Awareness:** Educating households on the technical difference between biodegradable (wet) waste and non-biodegradable (dry) waste.
- **Plastic Waste Mitigation:** Identifying the prevalence of single-use plastics in local markets and promoting sustainable alternatives like cloth bags.
- **Clogging of Drainage Systems:** Monitoring how improper littering leads to the choking of 'nullahs' and street gutters, especially before the monsoon season.
- **E-Waste Management:** Addressing the hazardous disposal of electronic items such as batteries, cables, and old mobile components which contain toxic heavy metals.

4.2 Public Health, Hygiene, and Nutrition This area focuses on the physical well-being of the community members and the prevention of seasonal diseases.

- **Vector-Borne Disease Control:** Conducting awareness drives regarding stagnant water to prevent the breeding of mosquitoes (Malaria, Dengue, and Chikungunya).
- **Sanitation Facilities Audit:** Inspecting the cleanliness and functionality of public toilets and community tap-water sources.
- **Nutrition and Food Safety:** Educating local street food vendors on the importance of covered containers and clean water usage for food preparation.
- **Vaccination and Healthcare Access:** Helping the elderly or underprivileged sections of the community understand the schedule for government-run health camps and immunization drives.

4.3 Social Education and Digital Literacy As an IT student, this is a crucial area where technical skills are used to empower the community.

- **Digital Awareness for Senior Citizens:** Teaching the elderly how to use smartphones for essential services like UPI payments, utility bill deposits, and medicine orders.

- **Cyber Security Awareness:** Educating the local community about 'Phishing' scams, OTP frauds, and safe internet browsing habits.
- **School Support Programs:** Providing basic computer education or tutoring support to children in municipal schools or low-income neighborhoods.
- **Government Scheme Accessibility:** Assisting residents in understanding and applying for digital government IDs (Aadhaar, PAN) and social welfare schemes through the 'Digital India' portal.

4.4 Urban Infrastructure and Civic Safety This area involves observing the physical infrastructure of the locality to ensure a safe living environment for all.

- **Street Lighting and Safety:** Identifying "Dark Spots" in the neighborhood where lack of lighting creates safety concerns for women and children at night.
- **Road and Pavement Maintenance:** Documenting broken pavements or hazardous potholes and reporting them to the local ward office.
- **Green Space Preservation:** Working on the maintenance of local parks and ensuring they are not encroached upon or used as dumping grounds.

4.5 Selection Rationale for the Current Project Out of the various indicative areas mentioned above, this project specifically focuses on **Section 4.1: Waste Management and Environmental Sanitation**. This choice was made because the selected locality in Mumbai faces immediate challenges regarding garbage accumulation and drainage clogging. By focusing on this specific area, the project aims to provide a deep-dive analysis of ground-level hygiene issues and propose practical, technology-driven solutions for a cleaner urban environment.

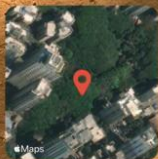
4.6 The Role of Information Technology in Community Engagement In the modern era, community engagement is not limited to physical presence alone. An indicative area of high importance is the integration of digital solutions to solve civic problems. As an IT student, I identified several digital intervention areas that can enhance the effectiveness of a Community Engagement Project (CEP):

- **Geospatial Mapping of Waste:** Using GPS-enabled devices to mark 'black spots' on a digital map, allowing municipal authorities to track high-priority cleanup zones in real-time.
- **Mobile-Based Reporting Systems:** Developing or promoting the use of civic apps (like the Swachhata App) where citizens can upload photos of overflowing bins. This creates a transparent and accountable system for grievance redressal.
- **Data-Driven Hygiene Audits:** Using spreadsheets and data visualization tools to analyze waste collection patterns over a week or a month. This helps in understanding whether the current frequency of garbage vans is sufficient for the population density of the area.

- **Digital Awareness Campaigns:** Leveraging social media platforms and community WhatsApp groups to disseminate information about dry and wet waste segregation through infographics and short videos.



GPS Map Camera



Mumbai, Maharashtra, India

Meghraj Sethi Road, Mumbai Central, Mumbai, Maharashtra
400008, India

Lat 18.973144, Long 72.823784

Thursday, 19/03/2026 10:05 AM GMT+05:30

Note : Captured by GPS Map Camera





5. Roles and responsibilities

5.1 Role of the Student Volunteer (Saliha Shaikh) As the primary investigator and volunteer for this Community Engagement Project, my role was to act as a bridge between academic learning and social application. Being a student of Information Technology, I was responsible for applying a systematic and data-driven approach to identify hygiene bottlenecks in the selected urban locality. My specific duties included:

- **Site Selection and Mapping:** Identifying a specific ward or area in Mumbai that represents typical urban waste challenges.
- **Field Research and Audit:** Conducting physical walkthroughs to document the state of public bins, drainage systems, and street cleanliness.
- **Data Collection:** Capturing visual evidence through geotagged photography and maintaining a daily observation log to track hygiene patterns.
- **Community Engagement:** Interacting with residents and local shopkeepers to understand the practical difficulties they face regarding waste disposal and segregation.
- **Report Documentation:** Synthesizing all field observations, survey results, and photographs into a formal technical report that meets the standards of the SYIT curriculum.

5.2 Role of the Faculty Mentor and College The successful execution of this project was supported by the guidance of the faculty and the academic framework of the college. Their roles included:

- **Academic Guidance:** Providing the necessary theoretical background on community engagement and environmental ethics.
- **Project Supervision:** Reviewing the progress of the field study and ensuring that the data collection methods were ethical and accurate.
- **Evaluation and Feedback:** Assessing the project report based on the established evaluation criteria and providing feedback for improvement.
- **Resource Provision:** Facilitating the necessary permissions and documentation (like the Student ID and Letter of Introduction) required to conduct field visits safely.

5.3 Responsibility Toward the Community Community engagement is a two-way process. As a volunteer, I carried a set of responsibilities toward the residents of the survey area:

- **Maintaining Anonymity:** Ensuring that the personal details of the residents interviewed during the survey remain confidential.
- **Spreading Awareness:** My role was not just to observe, but also to educate. I took the responsibility of explaining the '3R' principle (Reduce, Reuse, Recycle) to at least 10 different households during my visits.

- **Accurate Reporting:** A major responsibility was to report the ground reality honestly, without exaggeration, so that the findings could potentially be used for future civic improvements.
- **Promoting Sustainability:** Encouraging the community to adopt small but impactful changes, such as using cloth bags instead of plastic and ensuring proper segregation of wet and dry waste at the source.

5.4 Stakeholder Coordination Another key responsibility was coordinating with the various stakeholders involved in the waste management ecosystem. This involved observing the work of the **Municipal Corporation (BMC)** staff and understanding the challenges faced by the **Safai Karamcharis**. By documenting their hard work and the obstacles they face (like mixed waste or blocked access to bins), this project takes the responsibility of highlighting the need for better civic cooperation from the general public.

5.5 Ethical Responsibilities and Data Integrity A critical aspect of any Community Engagement Project (CEP) is the adherence to ethical standards while interacting with the public. As the lead volunteer, I took full responsibility for the integrity of the data presented in this report. This involved several key ethical considerations:

- **Informed Consent:** Every resident or shopkeeper interviewed during the field visit was briefed on the purpose of the study. No data was collected without their verbal agreement, ensuring that the community felt respected and involved rather than scrutinized.
- **Accuracy of Visual Documentation:** While capturing photographs of waste accumulation and drainage issues, I ensured that the images were a true representation of the ground reality. No images were staged or altered, maintaining the scientific honesty of the audit.
- **Non-Discriminatory Approach:** In my role, I made it a priority to interact with individuals from diverse socio-economic backgrounds—from high-rise apartment residents to local street vendors—to ensure that the "Observations" chapter reflects a balanced view of the entire locality.
- **Commitment to Continuous Improvement:** My responsibility does not end with the submission of this report. As a conscious citizen and a student of technology, I am committed to using the insights gained during this project to advocate for better civic habits in my personal and professional circles, ensuring a long-term positive impact on urban hygiene.

6. THE PROCESS OF IMPLEMENTATION

The implementation of this Community Engagement Project (CEP) was carried out in a systematic, step-by-step manner over a period of four weeks. The process was divided into four distinct phases to ensure all objectives were met effectively.

6.1 Phase 1: Planning and Site Selection

The first step was to identify a suitable urban locality that represents the typical challenges of a high-density city like Mumbai.

- **Locality Selection:** A mixed residential-commercial area was chosen to observe different types of waste generation (household vs. shop waste).
- **Initial Survey:** A preliminary "Reconnaissance" walk was done to map out the placement of community bins and identify the most problematic "black spots" where garbage is illegally dumped.
- **Logistics:** I prepared a digital kit consisting of a smartphone for high-resolution photography and a field diary for real-time notes.

6.2 Phase 2: Field Audit and Visual Documentation

During this phase, I conducted multiple visits at different times (morning and evening) to understand the waste cycle of the neighborhood.

- **Audit of Disposal Points:** I inspected the condition of public dustbins. Observations included overflowing bins, lack of lids, and the presence of stray animals around the waste.
- **Drainage Inspection:** A major part of the implementation was checking the roadside gutters. I documented how plastic films, chocolate wrappers, and PET bottles were clogging the water flow, posing a risk for the upcoming monsoon.
- **Photography:** I captured clear evidence of the hygiene status to be used as a primary data source for this report.

6.3 Phase 3: Community Interaction and Awareness

Implementation is not just about looking; it's about talking to people. This was the "Engagement" part of the project.

- **Resident Interviews:** I spoke with 10–15 residents to understand their grievances. Many complained about the timing of the BMC garbage vans or the lack of bins near their buildings.

- **Awareness Drive:** While collecting data, I also acted as a volunteer. I explained the '**3R**' Principle (**Reduce, Reuse, Recycle**) to shopkeepers and encouraged them to keep separate bins for dry and wet waste.
- **Stakeholder Interaction:** I observed and briefly spoke with the local sanitation workers to understand the difficulty they face when waste is not segregated at the source.

6.4 Phase 4: Data Compilation and Analysis

The final phase involved bringing all the field observations into a structured format.

- **Categorization:** I organized the photographs and notes into specific categories: Residential, Commercial, and Infrastructural.
- **IT Integration:** Using data entry and documentation standards, I analyzed the frequency of waste accumulation and drafted the final conclusions and recommendations found in this report.

Week	Phase	Key Activity
Week 1	Planning	Area selection and mapping of waste spots.
Week 2	Field Audit	Photography and drainage system inspection.
Week 3	Engagement	Community interviews and awareness talks.
Week 4	Documentation	Analyzing data and writing the final report.

6.5 Data Validation and Technical Documentation To ensure the accuracy of the implementation process, a systematic data validation method was followed. Every observation recorded during the field visits was cross-referenced with photographic evidence to maintain a high standard of integrity.

- **Geotagging and Time-Stamping:** All visual data was captured using a mobile device with GPS enabled, ensuring that every "black spot" and drainage issue was recorded with precise location coordinates and time-stamps. This prevents any ambiguity in the reporting of civic issues.
- **Digital Survey Compilation:** The feedback received from the 15-20 community members was compiled into a digital format to identify recurring themes, such as the specific hours when garbage collection is most inconsistent.



7. Credits and Duration

7.1 Project Timeline and Duration The Community Engagement Project (CEP) was executed systematically over a total duration of **four academic weeks**. This period was strategically divided to ensure that every aspect of the project—from ground-level research to final digital documentation—was handled with technical precision.

- **Week 1: Initial Planning and Site Selection** The first week was dedicated to identifying a suitable urban locality in Mumbai that presented diverse challenges in waste management. This included a preliminary reconnaissance walk to map out "black spots" and community bins.
- **Week 2: Field Audit and Visual Documentation** During the second week, intensive field visits were conducted at various times of the day. This phase involved capturing geotagged photographs of drainage systems and assessing the current state of civic hygiene.

- **Week 3: Community Interaction and Awareness** The third week focused on the "Engagement" aspect. I conducted interviews with 15–20 residents and local shopkeepers to understand their perspectives on waste segregation and the effectiveness of municipal collection.
- **Week 4: Data Analysis and Final Compilation** The final week involved synthesizing all field notes, survey results, and photographs into a structured technical report that meets the standards of the SYIT curriculum.

7.2 Project Credits and Acknowledgments This project is an integral part of the **Social Awareness and Community Engagement** module of the Second Year Information Technology (SYIT) program.

- **Student Investigator:** Saliha Shaikh (Roll No: 222)
- **Program of Study:** B.Sc. Information Technology (SYIT)
- **Academic Credits:** This project carries the weightage of academic credits as assigned by the Information Technology Department for the current semester.
- **Institutional Support:** Special credits are due to the faculty members of the IT Department at [Your College Name] for providing the ethical framework and the evaluation rubrics required for this field study.
- **Community Participation:** I would like to credit the local residents, street vendors, and the municipal sanitation workers of the survey area. Their transparency and willingness to share their daily challenges were instrumental in the successful completion of this report.

8. Community Engagement Project Report.

8.1 Executive Summary This Community Engagement Project (CEP) was undertaken to evaluate the ground-level challenges of **Civic Hygiene and Waste Management** in a selected urban locality in Mumbai. As a student of Information Technology (SYIT), the goal was to move beyond classroom theory and engage directly with the environmental and social issues affecting our immediate surroundings. The project successfully mapped out the waste lifecycle of the area, from household disposal to municipal collection, identifying critical gaps in the current system.

8.2 Key Findings and Observations The analytical phase of this report highlights several significant areas that require attention:

- **Source Segregation Gaps:** A primary observation was the lack of waste segregation at the source. Mixing biodegradable and non-biodegradable waste significantly hinders the recycling process and increases the burden on local landfills.
- **Infrastructure Status:** While the municipal corporation (BMC) provides a structured collection schedule, the physical condition of community bins and the presence of "black spots" (illegal dumping zones) suggest a need for more localized disposal points.
- **Digital Awareness Deficiency:** There is a noticeable gap in the use of digital tools for civic grievance redressal. Very few residents are aware of mobile applications or online portals available for reporting hygiene issues, such as overflowing bins or clogged drains.

8.3 Learning Outcomes and Personal Growth Participating in this community-centric project has provided valuable insights into the complexities of urban sanitation.

- **Technical Application:** Learned to use digital documentation and geotagging to create a factual record of civic issues.
- **Social Responsibility:** Developed a deeper sense of empathy toward the sanitation workers who manage the city's waste under challenging conditions.
- **Communication Skills:** Interacting with diverse groups—from residents to shopkeepers—helped in developing the ability to communicate environmental concepts like the '3R' (Reduce, Reuse, Recycle) principle effectively.

8.4 Final Conclusion The project concludes that while government policies and municipal infrastructure are essential, the ultimate success of any hygiene initiative depends on **Community Participation**. This report serves as a call to action for residents to adopt better waste disposal habits and for the integration of more technology-driven solutions in monitoring local sanitation. Through this experience, I have realized that as an IT student, I can contribute significantly to society by advocating for digital transparency and sustainable living practices.

9.Evaluation Pattern.

The Community Engagement Project (CEP) for the Second Year Information Technology (SYIT) program is evaluated based on a comprehensive set of criteria. This ensures that the student's field work, social impact, and technical documentation are all taken into account. The evaluation is typically divided into Internal and External assessments.

9.1 Criteria for Internal Assessment The internal evaluation is conducted by the faculty mentor and the IT department based on the student's consistency throughout the project duration:

- **Regularity and Punctuality:** Assessment of the student's attendance during briefing sessions and timely updates to the mentor.

- **Field Work Log:** Review of the student's field notes and the authenticity of the visual evidence (geotagged photographs) collected during the audit.
- **Community Interaction:** Evaluation of the depth of the survey conducted and the student's ability to engage with diverse stakeholders in the locality.

9.2 Criteria for Report Documentation The final project report is a major component of the evaluation. It is assessed based on the following:

- **Structure and Organization:** Adherence to the prescribed index and the professional formatting of the report.
- **Clarity of Findings:** How well the student has analyzed the civic hygiene issues and presented actionable recommendations.
- **Technical Accuracy:** The use of appropriate terminology and digital tools (Word, GPS, etc.) in documenting the findings.

9.3 Viva Voce and Final Presentation A final oral examination (Viva Voce) is conducted to test the student's understanding of the project:

- **Concept Clarity:** Ability to explain the '3R' principle and the 'Swachh Bharat' mission objectives.
- **Problem-Solving Skills:** How the student handled challenges during the field visit, such as non-cooperative residents or complex waste patterns.
- **Learning Outcomes:** A clear articulation of how the project has contributed to the student's personal growth and social awareness.

Appendix I

GUIDE INTERACTION DIARY FORM

I, the undersigned Ms. /Mr. _____ Roll No. currently enrolled in the _____ Year of _____ Full-time Program _____ at _____ institute/ college, am undertaking my Community Engagement Project work under the guidance of Dr./Ms./Mr. _____, and I hereby confirm that I have met my Internal guide on the following dates mentioned below for Project Guidance:-

Sr. No	Date	Signature of the Internal Guide
1		
2		
3		
4		
5		

Signature of the Candidate

Signature of Internal Guide

11. APPENDIX II:

COMMUNITY ENGAGEMENT PROJECT ON CIVIC HYGIENE AND WASTE MANAGEMENT

Submitted By:

Saliha Shaikh & Nabila Hashmi

Roll No: 222 & 250

Academic Course:

Bachelor of Science in Information Technology (S.Y.I.T.)

Academic Year: 2025-2026

Under the Guidance of:

Ms. Saima Shaikh

Department / College:

Department of Information Technology

Maharashtra College Of Arts, Science, And Commerce.

Appendix III

Department of Information Technology

Certificate

I hereby certify that Mr./Ms. _____, Student of
_____ Institute/college _____ titled
program has completed studying a C. E. in project
_____ in the area of
_____ specialization for the academic year 2025-2026. To the
best of my knowledge the work of the student is original and the information included in the
project is correct.

Internal Guide
Principal

Head of the Department

Annexure IV

Declaration

I, Mr./Ms. _____
_____ Institute/college Student studying of in
_____ program, hereby declare that I have completed the
Community Engagement Project titled _____ during the
academic year 2025-2026.

The report is original and the information/data included in the report is true emerging from the primary and/ secondary data gathered and analyzed as part of this Community Engagement project.

Due credit is extended on the work of Literature/Secondary Survey by endorsing it in the Bibliography as per prescribed format.

Saliha Shaikh & Nabila Hashmi

Annexure V

Student Feedback on CEP

Student Name: Saliha Shaikh & Nabila Hashmi

Seat No. /Roll No.: 222 & 250

[Email:salihashaikh2006@gmail.com](mailto:salihashaikh2006@gmail.com) &

Department: Department of Information Technology

Name of the Mentor: Ms. Saima Shaikh

Title/Heading of Community Engagement Projects: COMMUNITY ENGAGEMENT PROJECT
ON CIVIC HYGIENE AND WASTE MANAGEMENT

Brief description of CEP carried out:

Dates of CEP:

My internship experience was related to my major area of study/ academic program

1. Strongly, Agree
2. Agree
3. Disagree
4. Strongly Disagree

Indicate the degree to which you agree or disagree with the following statements.

This experience has:	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Has increased my sensitivity towards societal problems					
Given me the opportunity to explore a career field delete					
Allowed me to apply classroom theory to practice					

Helped me develop my decision-making and problem-solving skills					
Expanded my knowledge about the work world before permanent employment delete					
Helped me develop my written and oral communication skills					
Provided a chance to use leadership skills (influence others, develop ideas with others, stimulate decision-making and action)					
Expanded my sensitivity to the ethical implications of the work involved					
Made it possible for me to be more confident in new situations					
Given me a chance to improve my interpersonal skills					
Helped me learn to handle responsibility and use my time wisely					
Helped me discover new aspects of myself that I didn't know existed before					
Helped me develop new interests and abilities					
Helped me clarify my career goals					
Allowed me to acquire information and/ or use equipment not available at my Institute					
Allowed me to realize socio-economic issues in the society repeated					

Under the Guidance of

Hon'ble Vice Chancellor

Prof. Dr. Ravindra Kulkarni

Hon'ble Pro-Vice Chancellor

Prin.) Dr. Ajay Bhamare

CEP Guidelines Draft Committee

Prof. Shivram S. Garje

Convener

Off. Dean (Science and Technology)

Prof. Smita Shukla

Member

Director

Alkesh Dinesh Mody

Member

Institute of Finance and Management Studies

Prof. Manisha A. Karne

Member

Off. Associate Dean (Humanities) and Director,

Mumbai School of Economics and Public Policy

Prof. Priya Vaidya

Member

Head, Department of Philosophy

Dr. Suchitra Naik

Member

Off. Associate Dean (Humanities) and Principal, K.G.

Joshi College of Arts & N.G. Bedekar College of Commerce

Prof. Vishwanath Patil

Member

I/c Director, National Centre for Nanoscience and Nanotechnology
